Not Finished, But Abandoned

by

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ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— David C. Makinson (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

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ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,

2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

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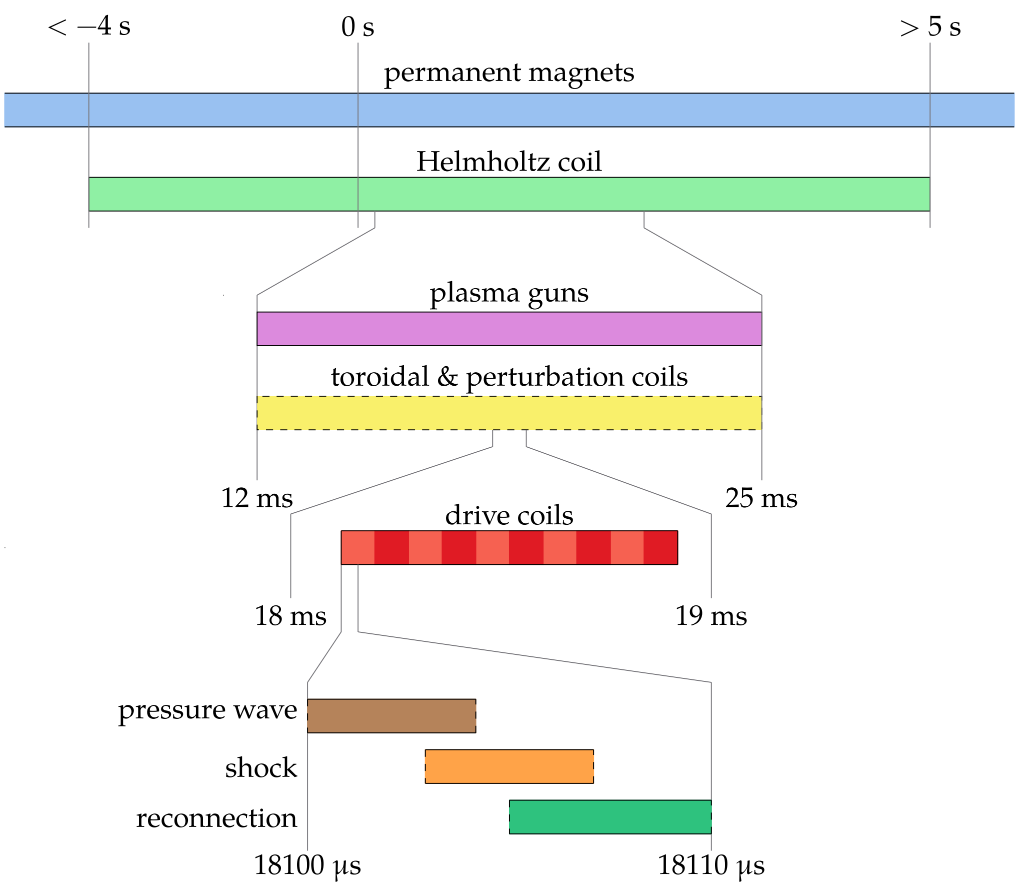

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3 MULTI-TIP LANGMUIR PROBE

3.1 What's a Langmuir probe

Hershkowitz: Hershkowitz (1989) Hutchinson: Hutchinson (2002) Olson: Olson (2020)

Plasma diagnostics make a broad phylogenetic tree ranging from the variety of plasma parameters they can measure to their size and complexity. In the Big Red Ball we prefer simplicity and cheapness to some of the wilder diagnostic varieties so a common probe that we use is a Langmuir probe.

3.1.1 Basic Construction

These Langmuir probes are quite basic in construction. At it's simplest, a conductor is placed in contact with the plasma and a voltage is applied relative to ground which we will call the probe voltage, V_{probe}. This will locally perturb the plasma and collect a current, I_{probe}, which can be seen in figure 3.1. If $V_{\mathrm{probe}} < V_{\mathrm{plasma}}$, an electric field is formed that repells electrons and attracts ions and the inverse for $V_{\mathrm{probe}} > V_{\mathrm{plasma}}$.

Langmuir probes have been developed to work in a very wide range

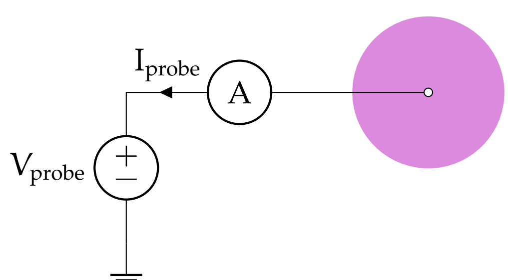


Figure 3.1: A circuit for a very simple Langmuir probe. The probe is biased to the probe voltage, V_{probe}, relative to ground by the power supply. The conductor (white circle) collects ions and electrons from the plasma (pink circle) causing a current to the probe, I_{probe}.

of plasma parameter configurations that well encompass the plasma parameters TREX achieves (see table 1.2).

3.1.2 Extracting Plasma Parameters

The most challenging part of these probes is the conversion of measured current to plasma parameters. Since we are trying to get a local measurement of the plasma parameters and we are also perturbing the plasma locally by touching it with a probe, we need to create a theory that correctly takes into account that perturbation such that we can get the plasma parameters. This often consists of a fitting procedure where the difference between the theoretical and experimental I-V curves is minimized.

Additionally, many a probe theory is purpose built for plasmas with no magnetic field. As we are studying magnetic reconnection in which there are in fact magnetic fields, this could change the I-V characteristic in unexpected ways. We'll discuss below why these effects are not very impactful and in future sections describe the theories we use for extracting the plasma parameters.

3.1.3 Plasma Perturbation

A problem often found in highly magnetized plasma devices is the long range perturbation to the plasma by the probe. In the Big Red Ball there are times where we have a relatively strong field and times when we don't so let's examine both cases.

3.1.3.1 Unmagnetized Perturbation

Let's examine this from an unmagnetized starting point. In this configuration, the probe collects nearly isotropically to an extent limited by the Debye length which creates a sheath around the probe and operates to shield the electric field caused by the difference between V_{probe} and V_{plasma}.

This length scale is very short and plasma can easily refill the sheath in this unmagnetized system.

3.1.3.2 Magnetic Field Effects

Now in a highly magnetized system (gyroradius far smaller than probe length scale) the plasma interacting with the probe comes from a flux tube approximately the size of the probe area projected into the magnetic field direction. Because of this, the probe can start "emptying out" the flux tube because of the difficulty of plasma to diffuse across field lines into the flux tube. This means the extend of the probe perturbation is far longer.

3.1.3.3 Perturbations in our System

We are quite lucky in that our system geometry reduces the impacts of flux tube emptying and the plasmas we are studying are often relatively unmagnetized.

3.1.3.3.1 Geometric Considerations The TREX experiment is (often) cylindrically symmetric and our probes are placed in different toroidal planes (see figure 2.6 for coordinate system reference). This means that without any toroidal field, any emptying only occurs in the plane of the probe which may still affect the probe I-V characteristic but is not affecting other probes measurements in a significant manner. When we do have a toroidal field, we also often have non-uniform B_r and B_z which means the flux tube passes through many different plasma parameters. This may help to average out any flux tube emptying because at different points along the flux tube more plasma may be able to enter the tube than just what is expected at the probe causing a reduced perturbation length.

3.1.3.3.2 Magnetic Field Considerations As we have magnetic fields in our system, we do need to worry about these flux tube effects. Fortunately, our magnetic field are relatively small and in the region we are often most interested in (where reconnection is occurring) the magnetic field is far smaller than the initial background field. We characterize this smallness by the ratio of the gyroradius to the radius of the probe area which is on the order of ??? (see 3.1).

3.2 Hardware

There has been a long and storied history of probe building on the Big Red Ball expecially for the development of multi-tip Langmuir probes. For some more information about our probes, see Olson (2020).

3.2.1 Physical Constraints

Our Langmuir probes need to satisfy a number of requirements.

- **Length**: The probe is attached to a port on the surface of the machine. This means it starts at a spherical radius of 1.5 m and must be inserted from there. Thus the insertable length should be fairly long and must be stable. This is described in 3.2.2.

- **Material**: We want our probe to survive the plasma exposure and not affect our vacuum pressure via offgassing. We also don't want to have to worry about secondary electron emission. We discuss this in 3.2.3.

- **Probe Size**: We want to measure the plasma locally so our probe body needs to be small relative to the plasma gradients. It also needs to be able to fit through ball joints that can be mounted to the machine that have a radius of ???. These considerations are taken into account in 3.2.3.

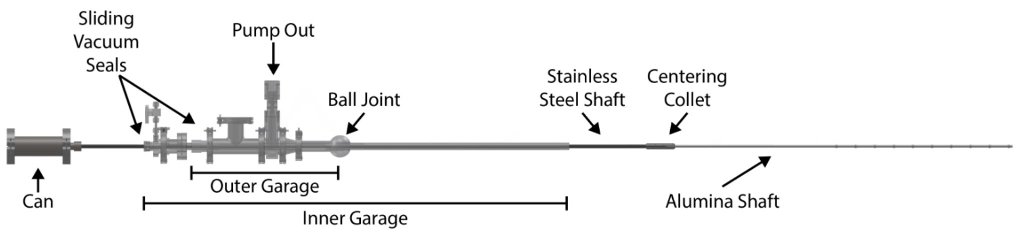


Figure 3.2: Three stage probe shaft with an outer and inner garage and the main stainless steel shaft that connects to the probe.

3.2.2 Shaft Components

For our probe to reach everywhere we want and also actually be in the position we desire we need a long shaft with a sufficient amount of support to be accurate to a few centimeters. Why a few centimeters? Our plasma dynamics are reproducible but not to such an extreme that a centimeter is noticeably different. Also it's difficult with the probe length to achieve too much better.

3.2.2.1 Garages

As can be seen in figure 3.2, the probe shaft is a multistage system. Each stage consists of some tubing where connections between different stages are made using quick-connect high-vacuum fittings (like part 3486N32 from McMaster-Carr).

We mount the probe to the machine using a flange on the outer garage or a ball joint (as seen in the figure). Then we can move each garage independently to insert the probe a desired length.

3.2.2.2 Shaft materials

We need to sufficiently support our probe and the shafts themselves so we use stainless steel until we enter the plasma region in which we change to alumina.

For the shafts, stainless steel is vacuum safe, tolerant to plasma interaction, and can support the weight of the structure easily without excessive flexing. Unfortunately, as a conductor, it will act as a long Langmuir probe that will create a ray in the machine at a single potential. This will perturb the plasma much more significantly than using an insulator.

To that end we use alumina, a high temperature ceramic who's surface will accumulate charge such that it'll be left at the floating potential (the potential at which the ion and electron currents balance). This helps to perturb the plasma less as there is no net current to the shaft and thus the plasma should (hopefully) be less perturbed.

Right near the end of the probe shaft, we use stainless steel tubing to connect from the alumina to the probe body. This is to reduce the overall size of the shaft in perturbing the plasma at the probe and is only for 10 cm of length.

3.2.2.3 Centering Collet

Each shaft/garage, is held in place by the sliding vacuum seal that connects it to it's larger garage. Unfortunately this gives only a single point of contact for making the shafts concentric. This would cause excessive drooping of the probe and may be so bad as to cause the probe to hit the edge of the machine when being inserted/removed.

To fix this issue, we have used centering collets that allow the inner shaft/garage to move while being held in place by the outer garage. These can be both machined or 3D printed. When 3D printed we can make these to also hold the o-rings so they are double-acting.

3.2.2.4 Feed-thru

We need some way to extract the signals from the probe tip. Because of the construction of our probe and for simplicity purposes, the probe shaft is at vacuum pressure. Thus we attach a feed-thru to the end of the can

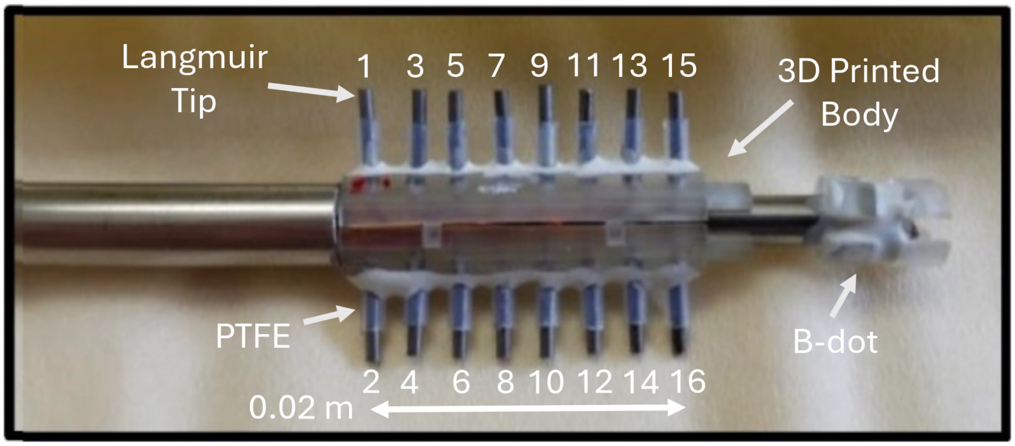


Figure 3.3: Picture of the end of the multi-tip Langmuir probe. Image from Gradney (2025).

from figure 3.2. This is constructed using two DB-25 hermetically sealed connectors where we solder the wires directly to the vacuum side of the can and then can easily move and connect the probe externally.

3.2.3 Tip Components

3.2.3.1 Probe body

Our probe does not need to withstand extreme heat fluxes and our vacuum conditions don't need to be perfect so we are lucky enough to be able to use a 3D printed probe body.

This is constructed using stereolithography which gives a sufficient resolution for the probe. Additionally the plastic used holds up well in vacuum. The body must connect to two different diameter stainless steel tubes (as can be seen in figure 3.3). Thus we use two annuli whose inner/outer radii match the tubing.

The tubing exists to connect the probe to the main shaft holding it up and also a small B-dot coil which is discussed in 3.2.4.

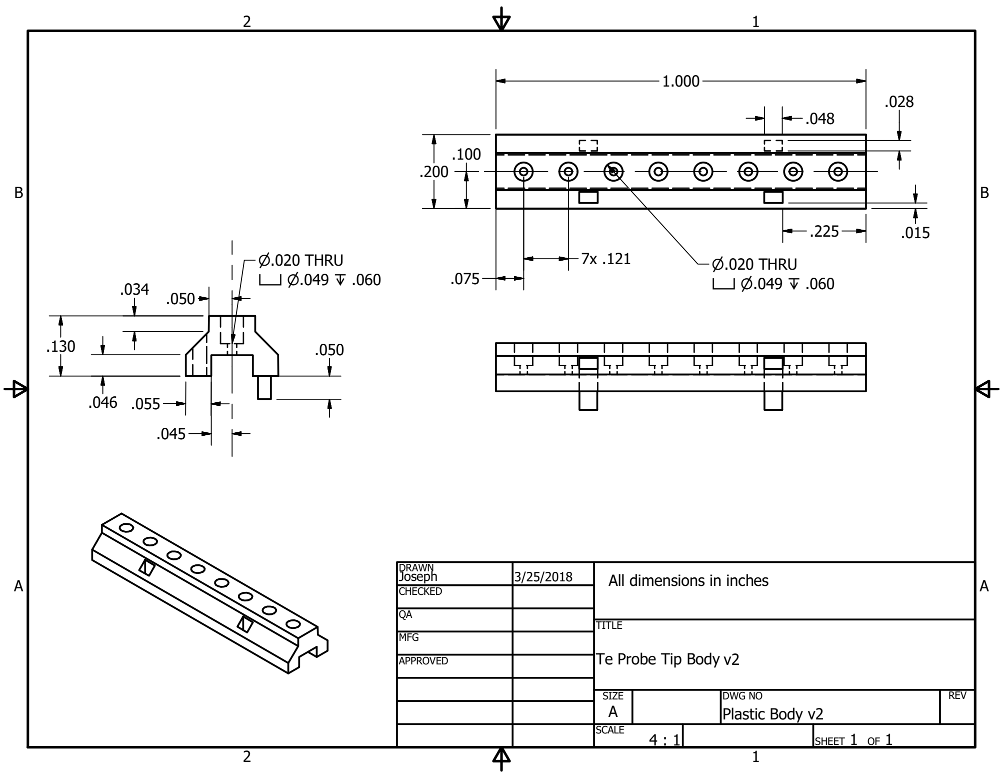


Figure 3.4: Probe body from engineering sketch. Created by Joe Olson.

3.2.3.2 Tip material + size

To collect current, we use 16 separate tips which each sample a point on the IV curve. We make these tips out of Molybdenum which requires us to connect the tips to the signal wire using a friction fit.

3.2.3.2.1 Why use Mo Even though Molybdenum has it's difficulties with connections, it has a high work function which helps to reduce secondary electron emission which can change the shape of the IV-characteristic.

3.2.3.2.2 Size considerations We expect the current to the tips to be on the order of 1 kA/m^2. We measure the current across a 'sense' resistor (as discussed thoroughly in 3.3.2) but we don't want to have to use high power resistors for this. Thus we choose our current to be on the order of 10 mA which is feasible by having a tip area on the order of 10 mm^2. This amount of current also gives a reasonable signal-to-noise ratio.

Unfortunately, the probe is not all that cylindrical which means that the idealized probe theories we use are not as precise as we may have hoped. Future probe designs could use longer and smaller diameter tips to avoid this issue.

3.2.3.3 Tip Shielding

We protect each tip from the base of the probe body using PTFE/teflon. Teflon isn't great in a vacuum environment because of offgassing but does the trick and is cheap and easy to use. After inserting the tip into the probe body we push the teflon on and apply a small amount of vacuum-safe epoxy at the base to hold the shielding in place.

3.2.3.3.1 Why shield at all? We could potentially shorten each tip so the collecting area extends all the way to the probe body. Unfortunately because of the increased size of the body this can cause shadowing effects where tips on one side of the probe have a different IV-characteristic than those on the other. This 'shadowing' is caused by plasma flow parallel to the tip direction whcih gets intercepted by the probe body.

3.2.3.3.2 First order mach effects to avoid Another reason to shield the probe is to avoid mach effects that would cause different IV-characteristics for tips on opposite sides. As can be seen in figure 3.5, in an ideal mach probe the ions are prevented from easily accessing the "shadowed" side of the probe (right side) because of their inertia. If we shrink the probe body and increase the distance the tip is from the probe the mach effect is reduced because ion inertia is not sufficient to prevent collection. This is

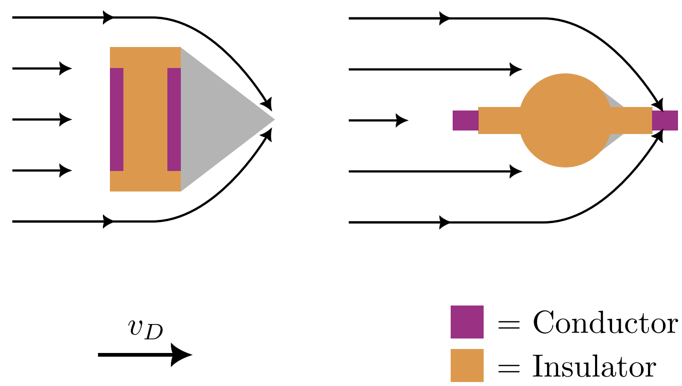


Figure 3.5: Simplified cartoon of the expected Mach effect for the multitip probe. The black arrows in the top figures are the approximate paths ions take due to the electric fields around the probe. The grey shaded region is where the ion density is significantly diminished. The left probe is a classic Mach probe geometry and the right probe is the multitip probe whose geometry is purpose built to avoid this Mach effect. Figure inspired by Merlino and D'Angelo (1987), Fig. 10.

only approximate though and we sometimes do see effects that could be caused by shadowing. For an in depth discussion of mach probes we refer the interested reader to Chung (2012).

3.2.4 B-dot Coils

Since we don't know the exact location of the probe and the plasma is not perfectly reproducable, it's especially useful to have measurements we can compare to other diagnostics to locate the probe in space and around the reconnection region. To do this we use hand wound three axis Bdot coils that can measure the change in magnetic field near the tips.

3.2.4.1 Size Considerations

Through past trial and error and many a day spent hand winding coils for past students, we've found a reasonable vector area magnitude to use for the B-dot coils. During reconnection the largest $\partial_t \mathbf{B}$ is on the order of 10 kT/s. Thus, from Faraday's law,

$$\mathcal{E} = \mathbf{A} \cdot \partial_t \mathbf{B} \tag{3.1}$$

where $\mathcal{E}$ is the electromotive force and $\mathbf{A}$ is the vector area of the coil Jackson (1999). This means that for a signal of 1 V we are looking for a cross-sectional area of 10^{-4} m^2.

Using a 3D-printed B-dot body (seen in right in figure 3.3) we wrap wire around a square shaped center six times giving an area of $\sim 5\times10^{-5}$ m^2. Although smaller than that required for a 1 V measurement, this makes sure the signal doesn't become too large should some kind of resonance exist in the circuit.

3.2.4.2 Noise Reduction

Since common mode noise can relatively easily couple to the coils, we use twisted pair for each wrapping direction. This gives us two circuits that have nearly opposite vector area, $\mathbf{A}_1 \approx -\mathbf{A}_2$, such that we can subtract the two measurements and thus remove common mode contributions. We will discuss this more in 3.3.3.4.

3.2.5 Length Scales

Table 3.1 contains all information that helps us decide on our Langmuir probe theory and signal interpretation. There are a few key features to note that let us validate the use of finite area B-dot coils and multiple tips.

Our ion and electron gyroradii are both relatively large compared to the probe tips and B-dot coils. This means that if we assume the plasma is relatively uniform in a flux tube, the entire probe is sampling a quite uniform plasma.

Name	Symbol	Equation	Value
Tip radius	r	—	5e-4
Tip length	l	—	2e-3
Neighboring tip separation	—	—	2e-3
Across probe tip separation	—	—	1.5e-2
B-dot coil side length	—	—	3e-3
Debye length	λ_D	$\sqrt{\frac{\epsilon_0 T_e}{ne}}$	3e-5
Ion Gyroradius	r_i	$\frac{m_i\sqrt{2m_i\bar{v}_i}}{eB}$	6e-1
Electron Gyroradius	r_e	$\frac{m_e\sqrt{2m_e\bar{v}_e}}{eB}$	1.5e-1
Ion mean free path	λ_i	$\frac{\bar{v}_i}{\nu_{i,i}}$	9e-4
Electron mean free path	λ_e	$\frac{\bar{v}_e}{\nu_{e,e}}$	4.5
Electron energy relaxation length	λ_ε	* Eq. 5	4.5
Sheath thickness	h	$\sim \lambda_D$	3e-5

Table 3.1: Length scales of the multitip probe. All lengths are in meters and are approximate. For the interested reader and for the energy relaxation length, please refer to Demidov et al. (2002) for Langmuir probe relevant physics of these scales. These parameters were calculated with $n = 10^{18}$ m^{-3}, $T_e = 20$ eV, $T_i = 0.1 \times T_e$, and $B = 10^{-4}$ T. For the mean free paths we have chosen the frequency that gives the shortest length.

3.3 Circuitry

The multitip Langmuir probe consists of a number of circuits that work together to give us some nice measurements. Here we enumerate their operation methods, upgrades, signal extraction, and calibration techniques.

3.3.1 Bias Source

We have thus far had two different circuits that we've used for biasing each tip relative to the laboratory ground. The first, made by Jan Egedal while at VTF is manually controlled, while the second, made by me is remotely operated. We'll discuss how they work and their pros and cons.

3.3.1.1 Manually Controlled

Our previous setup consisted of multiple electrolytic capacitors who's voltage is set by a series of resistors that split the voltage between the negative power supply (V_-), ground, and the postivie power supply (V_+).

3.3.1.1.1 Principles of operation Figure 3.6 shows a 4 tip biasing circuit (our circuit uses 16 tips but we have simplified it). The 'rails' (which are the maximum and minimum voltage of the circuit) are set by power sources at V_+ and V_- which are biased relative to the ground input connection labeled `Jgnd`. A series of resistors splits the voltage between the rails and we choose one of the tips to be at laboratory ground. This lets us set the maximum and minimum voltage along with setting how many tips are positively/negatively biased. Also, moving the ground changes the voltage separation between tips so we can get higher resolution. The voltage separation charges the respective capacitors for each tip to a bias voltage, $V_b =$ `Vb`. The capacitors exist to keep the bias voltage of each tip nearly constant during an experimental shot.

Now let's calculate the bias for each tip. Let $i, j \in \{1, \ldots, N\}$ be the index of the tip and the index of the ground connection respectively. Then,

$$V_{bi} = \begin{cases} \frac{V_+}{j}(j-i) & \text{if } j \geqslant i \\ \frac{V_-}{N+1-j}(i-j) & \text{if } j < i \end{cases} \tag{3.2}$$

where V_{bi} is the bias voltage for tip i.

3.3.1.1.2 The good, the bad, and the ugly Because of the simplicity of this system, it has proven to work relatively well since it's been built. Unfortunately since most of it's components are free-standing and the wired connections undergo some movement during travel and assembly, various solder joints are prone to breaking. Additionally because of the

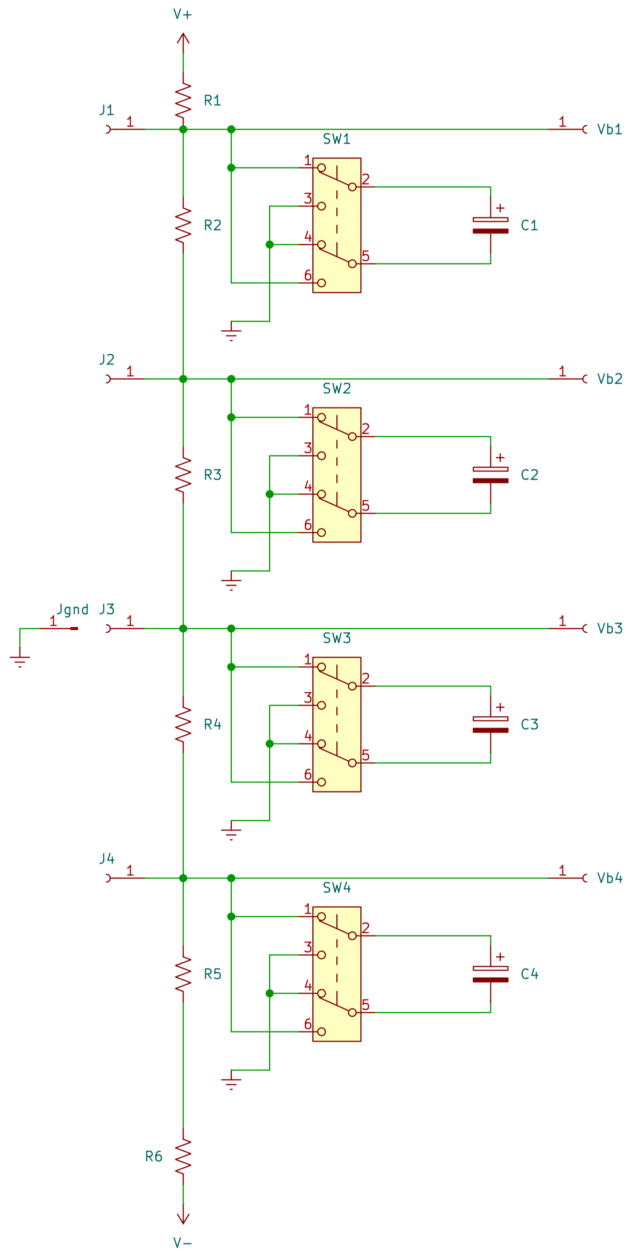


Figure 3.6: Old biasing circuit for the multitip Langmuir probe.

need to manually flip the polarity of each electrolytic capacitor using a switch mounted to a board and also move the ground via a banana connection, it is infeasible to remotely operate the bias supply. Additionally this circuit has no protection against setting the tip voltages extremely far apart which may cause arcing between wires or tips.

3.3.1.2 Remote Controlled

We've upgraded the previous setup by allowing the bias to be remotely controlled. This is done by removing the switches needed in the past setup and instead having one power supply set the voltage of the most negative tip and the other power supply set the maximum bias voltage difference between tips.

3.3.1.2.1 Principles of operation This system operates by having a large capacitor that is set to some negative voltage relative to ground. Then, all the other individual capacitors for each tip has it's negative terminal set to the output of the large capacitor. Thus we can change the bias of all the tips by the same amount by changing the negative bias. The splitting of the bias voltage is similar to the previous setup where the voltage from the positive power supply is split using a series of resistors. The circuit schematic and images of the circuit can be seen in figures 3.7 and 3.8 respectively.

We can write out the individual tip biases mathematically as

$$V_{bi} = \frac{V_+}{N+1} i + V_- \tag{3.3}$$

where V_{bi} is the tip bias, $V_+ \geqslant 0$ and $V_- \leqslant 0$ are the applied biases, i is the tip number, and N is the number of tips.

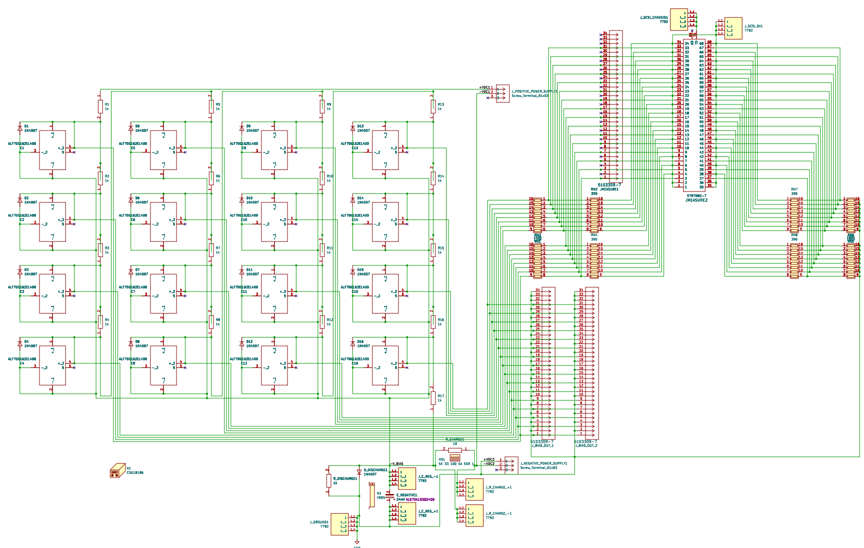
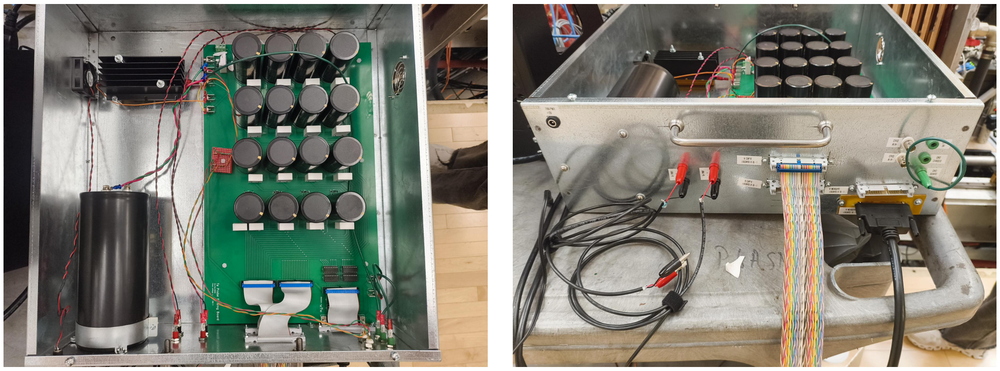


Figure 3.7: Circuit diagram of the new biasing setup. The left side consists of capacitors with reverse voltage protection diodes. Each capacitor is separated from the next by a resistor. At the bottom is the large capacitor for setting the negative voltage and it's diode & discharge resistor. The top right are connections necessary for dividing down the bias voltage for measurement and the bottom left are ribbon cable connections for applying voltage to the tips.

3.3.1.2.2 Problems Although this system works wonderfully to remotely control the bias such that we don't have to be running up and down the stairs all day long, it does come with it's own set of issues.

During running when the probe was at low radius and $V_{-} = -225$ V, the positive power supply broke and a short to ground was formed internal to the power supply between the negative and ground bananda connection terminals. This was most likely due to some transient that sent the positive power supply more than 300 V from ground which the power



(a) Top view (b) Front view

Figure 3.8: An image of the circuitry for the new bias power supply. In panel 3.8a: the top left is a fan for cooling the charging resistor hidden beneath a heat sink in black, the bottom left shows the negative capacitor that sets the ground of the rest of the circuit, the top right black cylinders are the individual tip capacitors and white rectangles are high power resistors for the voltage division, bottom right are connections to the front panel. In panel 3.8b: the top left is the fan power, the red and black banana connections are to the power supplies, the rainbow ribbon cable is the connection to send bias to the tip circuit, the bottom right black cable is for measurement of the bias voltage, and the green banana is a connection to the laboratory ground.

supply can not do.

Additionally the 10 W discharge resistor initially used for discharging the negative capacitor burned out because the power load was higher than expected. We replaced it with a large carborundum resistor that can handle 50 W.

3.3.1.2.3 Calibration Calibration of the supply consists of two things:

- The power supplies are remotely controlled through LabView which communicates with a CompactRIO analog output module that sends an isolated control voltage to the power supply. This output signal has a voltage offset and is not a perfect conversion between set voltage

Name	Symbol	Value	Unit
Old Bias Circuit			
Positive bias voltage	V_+	$0 \leqslant V_+ \leqslant 250$	V
Negative bias voltage	V_-	$-250 \leqslant V_- \leqslant 0$	V
Tip capacitor		1.5	mF
Divider Resistor		???	Ω
New Bias Circuit			
Positive bias voltage	V_+	$0 \leqslant V_+ \leqslant 250$	V
Negative bias voltage	V_-	$-250 \leqslant V_- \leqslant 0$	V
Tip capacitor		1.6	mF
Divider Resistor		1	kΩ
Negative bias capacitor		24	mF
Charging resistor		10	Ω
Discharging resistor		1	kΩ

Table 3.2: Bias circuit component values

and output voltage that can be corrected by a multiplicative factor. We can do both these corrections by setting a voltage in LabView and seeing what voltage the power supply is outputting. This is just a problem of fitting a straight line with an offset.

- We also need to calibrate the measurement of the bias voltage. Since the digitizers have an internal offset voltage and there is a resistor divider between bias and measurement we must correct for both of those. This is similarly a case of measuring the output bias voltage and correcting the digitizer measurement to match that value.

3.3.2 Probe Circuit

Now that we can apply a voltage to the tip, we need some way to measure the current to the probe. We do this using a commonly used circuit setup with a few additions for our setup which can be seen in figure 3.9.

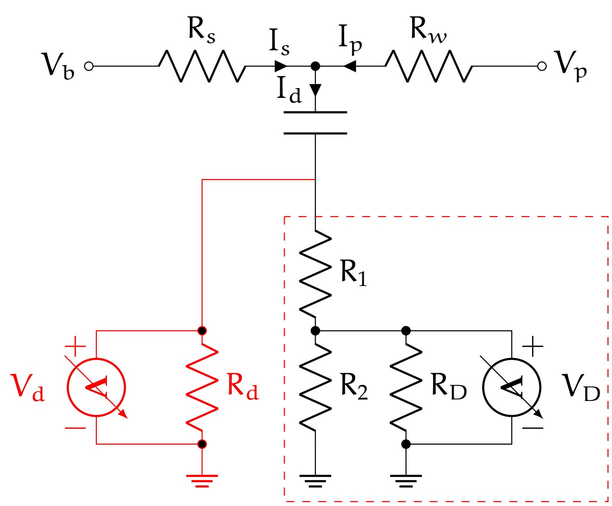


Figure 3.9: Circuit for a single tip on the multitip probe. We can reduce the circuit contained in the red dashed rectangle to the red circuit using the equation for a voltage divider.

3.3.2.1 Principles of operation

We apply the bias at the node labeled V_b. When the probe pulls current from the plasma, I_p, this changes the tip voltage and pulls the voltage closer to the floating potential since there is a voltage drop across the sense resistor, R_s. The capacitor and resistor divider RC times acts to shield the DC component of the probe bias voltage. The voltage difference across the sense resistor is measured by a digitizer which sees V_D but that resistor divider can be simplified to a measured voltage of V_d. We want the probe voltage, V_p, and the probe current, I_p, to fit the plasma parameters.

3.3.2.2 Single-ended measurement

We've used the TREX cells in the past which are single ended digitizers that can sample between 2.5 MHz (32 channels) and 20 MHz (4 channels). These work pretty well for our lower frequency experiments like those with the drive coils. Unfortunately because of the high voltages and large

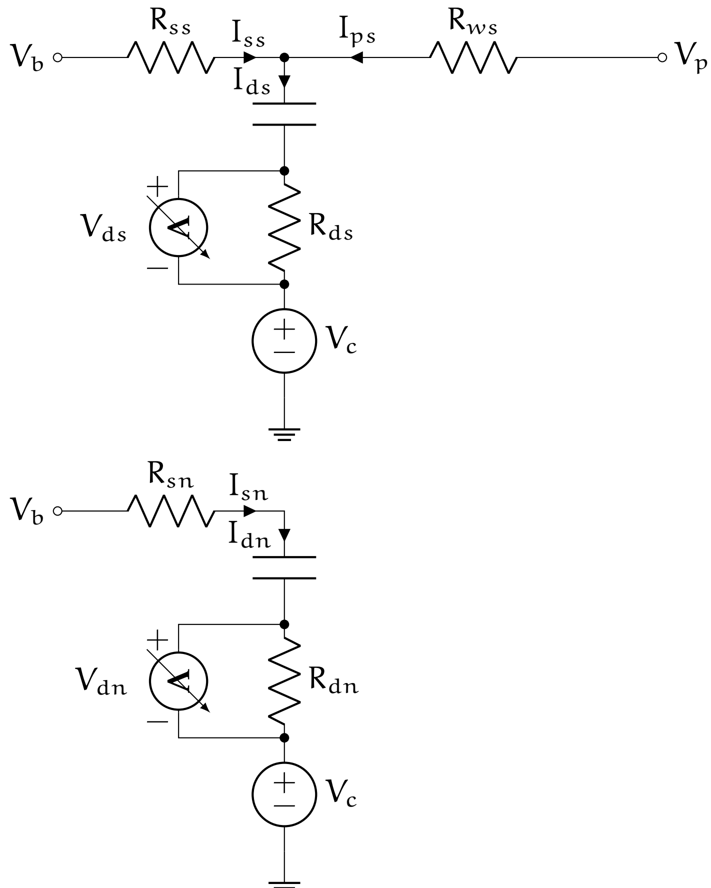


Figure 3.10: Circuit for a single tip on the Langmuir probe alongside the circuit for measuring noise. The V_b inputs are the same for both circuits but since the bias not change during a shot we will ignore any currents flowing between the circuits. The subscripts are organized as (element, circuit) where the elements are b = bias, s = sense resistor, w = wire, p = probe, d = digitizer and the circuits are n = noise, s = signal. In our equations we don't use the raw voltages from the digitizers (V_{ds} and V_{dn}) and instead use the currents (I_{ds} and I_{dn}) which we can find using $V = IR$. The noise is modeled as a changing chassis voltage.

Table 3.3: Probe circuit component values

currents, the ground of these digitizers can change a lot during a shot even if they are on isolation transformers to remove their reliance on the lab ground. This circuit can be seen in figure 3.11.

3.3.2.3 Differential measurement

We've developed a new circuit board to make use of our D-TACQ digitizers which measure using a differential input. Instead of measuring between the digitizer ground and the resistor divider output, we instead duplicate all the circuit elements on the board meaning each tip has two sense resistors, two capacitors, and two resistor dividers. One of the circuits is still connected to the probe tip through it's wire but the other is only connected to the bias voltage input. Thus we can measure the difference between these two signals.

This vastly improves the signal quality and helps to remove much of the noise we previously saw when the capacitor banks were triggered.

3.3.2.4 Circuit Solution

During a shot, noise is generated around the lab because of the high currents used to create the reversed magnetic field for reconnection. To remove this noise we use another circuit that has the same components but is not connected to a plasma facing tip which can be seen in figure 3.10. We'll call this other circuit the **noise circuit** and the circuit that is connected to the tip the **signal circuit**. If there was no noise in the lab we would expect to measure no fluctuations on the noise circuit digitizer.

In order to get the IV-curve needed for extracting out the plasma parameters we need to solve for the probe voltage and current. To do this, we'll take figure 3.10 as our model where we have modeled the noise

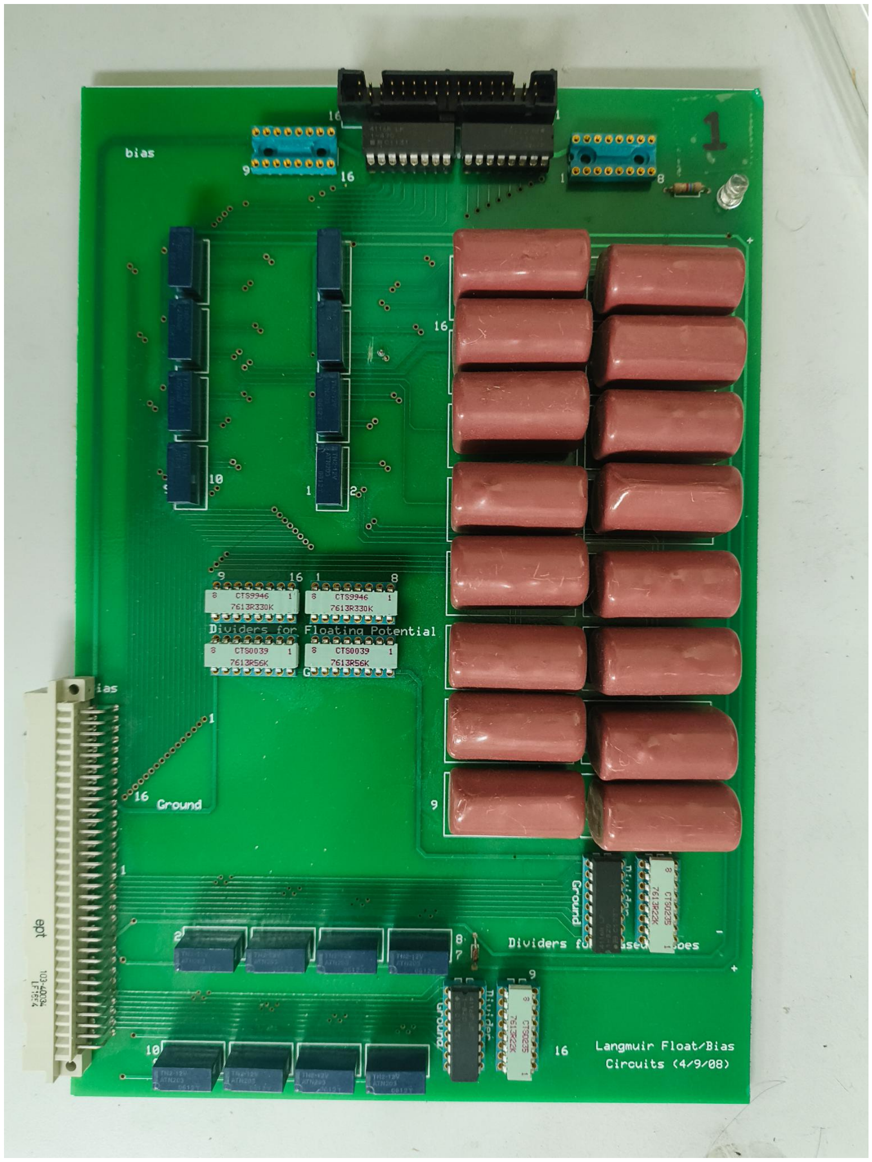


Figure 3.11: Image of circuit board used for single-ended measurements. The top ribbon connector is where the bias comes in. Just below that connector are multiple sense resistors. Often those are replaced with hand soldered higher power resistor networks. The pink rectangles on the right are the capacitors as can be seen in figure 3.9. The resistor networks in black and light blue in the bottom right are a resistor divider prior to measurement whose signals leave out the tan connector at bottom left.

Figure 3.12: Image of new circuit board

as a fluctuating ground voltage[1]. This isn't a perfect model as it ignores other sources of noise from the digitizer or bias fluctuations (which it may actually account for) but it works well for our purposes. The capacitor in this circuit has nearly no time dependence over the shot so we treat it as having a constant voltage drop of V_b.

3.3.2.4.1 How do we solve this Using Kirkoff's laws we can construct a system of linear equations. Kirkoff's current law gives us two equations for the nodes in the signal and noise circuits and Kirkoff's voltage law gives six equations connecting any end node combination in each circuit. We solve these equations using `SymPy`, a symbolic math library in Python. These give the following:

$$\begin{aligned}
I_{ps} =& \frac{1}{R_{ss}}[(R_{ds} + R_{ss})I_{ds} - (R_{dn} + R_{sn})I_{dn}] \\
V_{ps} =& \frac{1}{R_{ss}}[(R_{ds}R_{ss} + R_{ds}R_{ws} + R_{ss}R_{ws})I_{ds} \\
& - (R_{dn}R_{ss} + R_{dn}R_{ws} + R_{sn}R_{ss} + R_{sn}R_{ws})I_{dn}] + V_b
\end{aligned}$$

Since the digitizers are measuring the voltage output of the resistor divider we have replaced the measured voltages, V_D, with the currents, I_d, which is the total current flowing through the resistor divider. Since we know the measured voltage and the resistances of the divider and digitizer we can find the current using Fig. 3.9:

[1]We've also tried modeling the noise using a fluctuating current that appears on both the probe signal and where a probe would attach in the noise circuit. This did not work as well as the ground voltage fluctation method though.

$$
\begin{aligned}
R_d &= R_1 + (R_2^{-1} + R_D^{-1})^{-1}, \\
V_D &= \frac{(R_2^{-1} + R_D^{-1})^{-1}}{R_d} V_d, \\
I_d &= \frac{V_d}{R_d}
\end{aligned}
$$

Thus our final equations for the probe voltage and current are

$$
\begin{aligned}
I_d =& \frac{V_d}{R_d} = \frac{V_D}{\left(R_2^{-1} + R_D^{-1}\right)^{-1}} \\
I_{ps} =& \frac{1}{R_{ss}}[(R_{ds} + R_{ss})I_{ds} - (R_{dn} + R_{sn})I_{dn}] \\
V_{ps} =& \frac{1}{R_{ss}}[(R_{ds}R_{ss} + R_{ds}R_{ws} + R_{ss}R_{ws})I_{ds} \\
& - (R_{dn}R_{ss} + R_{dn}R_{ws} + R_{sn}R_{ss} + R_{sn}R_{ws})I_{dn}] + V_b
\end{aligned}
$$

3.3.2.4.2 Assumptions Here we have assumed that all the noise is coming from a fluctuating ground which is an alright assumption as we do expect significant ground fluctuations that could easily change the measurement. We see less of these fluctuations during shots with no plasma which may mean part of this noise is coming from current drawn by the probe. It remains to be seen if there are better ways to model the common-mode noise or if we are accidentally removing bias voltage fluctuations that may be necessary for better IV-characteristic reconstructions.

3.3.2.5 Calibration

Try as we might, we can't perfectly manufacture these probes such that the areas of each tip are the same and the various resistances all match. The best way to calibrate this probe is using the plasma itself.

3.3.2.5.1 Calibrating in Ion Saturation Regime When the current to the probe is dominated by the ions, the IV-characteristic is very flat. By setting all tips to the same very negative bias voltage we measure different currents to each tip. The relative difference between measurements is nearly the same as the relative area difference between those tips.

3.3.2.5.2 Calibrating on the exponential In the ion attracting regime but near the plasma potential, the shape of the IV-characteristic is approximately exponential. Since our initial plasma is very cold (~ 4 eV), the rise is very swift. Thus setting the tip potential to be near the plasma potential during the equilibrium plasma before the drive coils gives us the ability to calculate the wire resistance such that all the points on the IV-characteristic are at the same voltage.

3.3.2.5.3 Bayesian calibration We can actually do both of these calibrations using Bayesian statistics where we solve for the probability of the area (or resistance) based on the current to the probe. By finding the probability of the measured current for some area assuming some amount of noise and then applying a prior that forces the area to be in some resonable range let's us find the most probable area. The same can be done for the resistance.

3.3.2.5.4 Calibration using pressure wave Although we can calibrate the relative areas between probes we need some way to get the actual plasma parameters (not just the relative ones)! We do this using data from the previous chapter. Since that method works well to extract the initial density profile we can change the tip areas to best match those results.

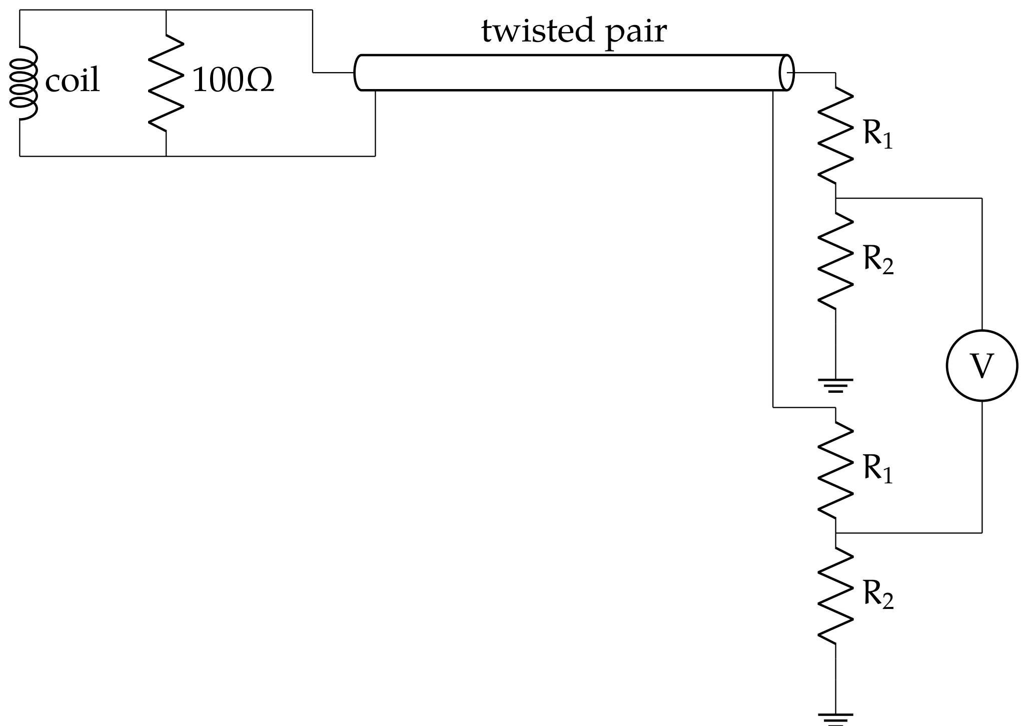


Figure 3.13: The Bdot coil circuit. The top left inductor and resistor and near the probe tip. The signals are extracted using twisted pair and then divided down before a differential measurement.

3.3.3 Bdot coils

The Bdot coils, while basic in theory, need some help to actually give physical results.

3.3.3.1 In-shaft circuit

Inside the stainless shaft connected to the probe, there is another circuit containing some resistors and various through holes. Since each coil wrap is really two coils (as they are wrapped using twisted pair) this is where those two coil signals are separated. The resistor is in place to improve the system response to high frequency.

3.3.3.2 Circuit model

We model the circuit using figure 3.13. The small resistor near the coil does not come into play for analyzing the signal. As of this writing we assume a linear response between fluctuating magnetic fields and measured voltage but a better solution would be measurement of the transfer function.

3.3.3.3 Calibration

Although we could use a handheld coil for calibrating the areas and directions of this coil, it is far easier and more accurate to use the drive coils as a calibration source. Without plasma, we can model the expected magnetic field assuming they induce current in the BRB walls such that no field can escape (this is true since the BRB shell is made of aluminum and the induced currents have continuous paths). Using these modeled fields we can match the measurement to the expected field and by rotating the probe and moving the probe figure out the vector areas.

3.3.3.4 Signal Interpretation

Once we have the vector area of each coil, $\mathbf{A}_i$, we can now extract the Bdot signal. According to Faraday's law,

$$\mathcal{E}_i = \mathbf{A}_i \cdot \dot{\mathbf{B}}. \tag{3.4}$$

Now we measure a signal, S_i, where

$$S_i = \frac{R_2}{R_1 + R_2} \mathcal{E}_i. \tag{3.5}$$

Assuming the coils make a basis for $\mathbb{R}^3$, we can turn this into an inverse problem:

$$\begin{bmatrix} S_1 \\ S_2 \\ \vdots \\ S_6 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_6 \end{bmatrix} \cdot \dot{\mathbf{B}} \tag{3.6}$$

$$\mathbf{S} = \mathbf{A} \cdot \dot{\mathbf{B}} \tag{3.7}$$

Now $\dot{\mathbf{B}}$ is overdefined since we have six measurements and three field components so using linear least squares[2] we can solve for $\dot{\mathbf{B}}$ and also the standard deviation of $\dot{\mathbf{B}}$. This is useful in understanding where we might expect error in our signals and whether we need to smooth signals to get rid of noise caused by the firing of the capacitor banks (which is sometimes needed).

3.4 Plasma Theory

The depth of theory needed to correctly understand Langmuir probes is near endless with a menagerie of exciting physical effects one could account for. I'm prone to dive a bit too deep into some of these various theories but after much literature review and some misteps we've come to a theory that seems to work relatively well for our results. Here I'll discuss the prior method using a very simple model and another method that gives much better results.

[2]We've also tried using generalized least squares (since coils made of the same twisted pair may have more correlated noise) but we did not see a significant improvement.

3.4.1 Simple Plasma Theory (Hutchinson)

Hutchinson (2002) develops a simple model for the current to a Langmuir probe that we have used for a long time. It assumes that the ion current is constant and the electron current follows an exponential and only looks at the electron repelling region.

3.4.1.1 Derivation

We won't reproduce the derivation here but will instead describe the physical interpretation.

The volume surrounding the probe is split into two regions: **the plasma** where the electric field is very small, quasineutrality holds, and the potential is near the plasma potential vs. **the sheath** where the ion electron charge densities are different and the electric field is largs such that the potential quickly jumps to the probe potential.

The ion current is just modeled by how many ions randomly cross the plasma-sheath interface (the interface will be denoted by subscript s). If the ions are cold (i.e. negligible initial kinetic energy) then they will only get to the probe via their random motion. The speed of these ions is set by the voltage at which the interface appears which is where

$$V_s = -\frac{T_e}{2e}. \tag{3.8}$$

V_s is the potential at the interface relative to the plasma potential. Assuming the electrons are Boltzmann and quasineutrality holds at the interface then

$$n_{is} = n_{es} = n_\infty \exp\left(\frac{eV_s}{T_e}\right). \tag{3.9}$$

Now we can calculate the ion current as

$$\begin{aligned} I_i &= A_s v_{is} n_{is} && (3.10)\\ &= A_s \sqrt{-\frac{2eV_s}{m_i}} n_\infty \exp\left(\frac{eV_s}{T_e}\right) && (3.11)\\ &= \exp\left(-\frac{1}{2}\right) A_s n_\infty \sqrt{\frac{T_e}{m_i}}. && (3.12) \end{aligned}$$

We often assume that the sheath area is the same as the probe area since the Debye length is so short.

For the electron current, we are assuming they are Boltzmann and they will hit the probe due to random motion. Since they are repelled by the probe the density will be smaller than in the bulk plasma. We can write the electron current as

$$I_e = A_p \frac{1}{4} \sqrt{\frac{2T_e}{m_e}} n_\infty \exp\left(\frac{eV_{\text{probe}}}{T_e}\right) \tag{3.13}$$

where the square root is the average electron velocity and the exponential is the factor the electron density is reduced.

The total current to the probe for $V_{\text{probe}} < V_{\text{plasma}}$ is thus

$$I = n_\infty e A_p \sqrt{T_e} m_i \left[\frac{1}{2}\sqrt{\frac{2m_i}{\pi m_e}} \exp\left(\frac{eV_{\text{probe}}}{T_e}\right) - \exp\left(-\frac{1}{2}\right)\right]. \tag{3.14}$$

3.4.1.2 Pros & Cons

This method has the benefit of being exceedingly simple to implement and often widely applicable. As long as the tips stay below the plasma potential this usually nicely matches our measurements. Unfortunately there's the catch, during a shot the plasma potential changes quickly by up to 100 or more volts. One option is to select only probe tips that look like they are below the plasma potential but this is both time consuming and prone to bias. Thus we look for a new method.

3.4.2 BRL

Laframboise (1966) created a model for the current to a cylindrical probe in a plasma with maxwellian ions and electrons that works for probe voltages above and below the plasma potential. Unfortunately, to my knowledge nobody has redone these calculations nor made a public repository with the code. To sidestep this issue we've used two simplifying approximations.

For probe potential greater than plasma potential, i.e. electrons attracted, we follow along with appendix F of Laframboise (1966). We assume the ions have zero temperature. Although we have no way currently to calculate the ion temperature it is an assumption we commonly make for plasmas from our plasma guns. Additionaly we assume we assume that the probe radius is far larger than the debye length. This approximation is valid for the initial plasma but can change throughout the shot. We also assume that all particles that touch the probe are absorbed by it.

Since the scale over which the potential changes is proportional to the debye length, we assume we have a planar probe. Thus Poisson's equation is given by

$$\frac{d^2\varphi}{dx^2} = -\frac{\rho}{\varepsilon} \tag{3.15}$$

Since the ions have zero temperature, they can not enter the $\varphi(x) > \varphi_0$ region and thus for the plasma charge density, $\rho(x)$, they can be ignored. Without loss of generality we'll set the plasma potential, φ_0, equal to 0. Since this region of electrons only plasma physically can't extend to infinity let's define $x = 0$ to be this boundary and let x increase towards the probe.

As this is a one-dimensional system then we have three conserved quantities. v_y and v_z which denote the transverse velocity will not be changed by the potential and the kinetic energy in the x direction along with the electrostatic potential, $E_x = mv_x^2/2 + e\varphi(x)$. e is the elementary charge. As all electrons will be absorbed by the probe, we will only have flow into the $\varphi(x) > 0$ region. Thus our distribution function looks like

$$f(x, E_x, E_t) = \begin{cases} n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{E_x(x)}{kT}\right) \exp\left(-\frac{E_t}{kT}\right) & v_x > 0 \\ 0 & v_x < 0 \end{cases} \tag{3.16}$$

We can now write the density as

$$\begin{aligned} n(x) =& \int_0^\infty dv_x \int_{-\infty}^\infty dv_y \int_{-\infty}^\infty dv_z f(x, E_x, E_t) \\ =& \int_0^\infty \frac{dv_x}{dE_x} dE_x \int_{-\infty}^\infty dv_y \int_{-\infty}^\infty dv_z f(x, E_x, E_t) \\ =& n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty \frac{\exp\left(-\frac{E_x(x)}{kT}\right)}{\sqrt{2m(E_x - e\varphi(x))}} dE_x \\ & \int_{-\infty}^\infty \exp\left(-\frac{mv_y^2}{2kT}\right) dv_y \int_{-\infty}^\infty \exp\left(-\frac{mv_z^2}{2kT}\right) dv_z \\ =& n_0 \sqrt{\frac{m}{2\pi kT}} \int_0^\infty \frac{\exp\left(-\frac{E_x(x)}{kT}\right)}{\sqrt{2m(E_x - e\varphi(x))}} dE_x \end{aligned}$$

Let's define $\eta = n(x)/n_0$ where n_0 is the number density at infinity. Let's also define $\chi(x) = e\varphi(x)/kT$ and $\beta = E_x/kT$ as the normalized potential and energy respectively. Thus we can write the normalized density as

$$\begin{aligned} \eta(x) &= \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{\exp(-\beta)}{\sqrt{\beta - \chi(x)}} d\beta \\ &= \frac{e^{-\chi(x)}}{2} (1 - \mathrm{erf}(\sqrt{-\chi(x)})) \end{aligned} \tag{3.17}$$

Using our now normalized quantities, let's plug them into Poisson's equation 3.15 to find

$$\begin{aligned}\frac{\varepsilon kT}{e}\frac{d^2\chi}{dx^2} &= e\eta(x)\\ \lambda_D^2\frac{d^2\chi}{dx^2} &= \eta(\chi(x))\\ \frac{d^2\chi}{ds^2} &= \eta(\chi(s))\end{aligned} \tag{3.18}$$

where $s = x/\lambda_D$ and $\lambda_D = \sqrt{\varepsilon kT/n_0 e}$ is the Debye length in the undisturbed plasma.

Since the potential is zero at the sheath boundary and the derivative is smooth, $\chi(0) = 0$ and $d_s\chi(0) = 0$. For all $s > 0$, the density of electrons should be non-zero, i.e. we should never achieve a full vacuum except at infinity. Thus,

$$\frac{d^2\chi}{ds^2} > 0, \ \frac{d\chi}{ds} = 0 \Rightarrow \frac{d\chi}{ds} > 0 \Rightarrow \left(\frac{d\chi}{ds}\right)^{-1} = \frac{ds}{d\chi}$$

This means

$$\begin{aligned}\frac{d^2\chi}{ds^2} &= \frac{d}{ds}\left(\frac{ds}{d\chi}\right)^{-1}\\ &= \frac{d\chi}{ds}\frac{d}{d\chi}\left(\frac{ds}{d\chi}\right)^{-1}\\ \eta(\chi) &= -\left(\frac{ds}{d\chi}\right)^{-3}\frac{d^2s}{d\chi^2}\\ \Rightarrow \int_0^{\chi}\eta(\chi')d\chi' &= \frac{1}{2}\left(\frac{ds}{d\chi}\right)^{-2}\\ \Rightarrow \frac{ds}{d\chi} &= \frac{1}{2\int_0^{\chi}\eta(\chi')d\chi'}\\ \Rightarrow s(\chi) &= \int_0^{\chi}\frac{d\chi'}{2\int_0^{\chi'}\eta(\chi'')d\chi''}\end{aligned}$$

This integral can be evaluated numerically. Thus we can pass a probe potential, χ, and after integrating get the distance from the probe to the start of the sheath.

Since all particles that enter the sheath are collected, from equation 3.16, then we need only find how the surface area of the sheath changes for different potentials. Letting I_0 be the current collected from the electrons when the probe is at the plasma potential then the current to the probe is

$$I(\chi) = I_0 \left(1 + \frac{\lambda_D}{R_p} s(\chi)\right)^2 \tag{3.19}$$

Now that we have a model for the current to the probe for the electron attracted region, we move on to modeling the current to the probe for the ion attracted region. There has been much more work put into this region from Steinbrüchel et al. Steinbrüchel (1990) Karamcheti and Steinbrüchel (1999), Mausbach Mausbach (1997), and Chen Chen (1965). Steinbrüchel fitted a model of the form

$$I = I_{i0} a(-\chi)^b + I_{e0} \exp(\chi) \quad \chi < 0$$

where a and b were fit to the data given by Laframboise. Mausbach used

$$I = \begin{cases} I_{i0} a_i (b_i - \chi)^{c_i} + I_{e0} \exp(\chi) & \chi < 0 \\ I_{e0} a_e (b_e + \chi)^{c_e} & \chi \geqslant 0 \end{cases}$$

where $a_{i,e}$, $b_{i,e}$, and $c_{i,e}$ are fitted parameters for the ions and electrons.

Unfortunately neither of these correctly predict the ion current to the probe for large χ so we instead use the parametrization given by Chen Chen (2001) which better predicts ion current for $\chi < 0$. The total current to the probe is given by

$$I = I_{i0} i(\chi) + I_{e0} \exp(\chi)$$
$$\frac{1}{i^4} = \frac{1}{(A\chi^B)^4} + \frac{1}{(C\chi^D)^4}$$

where A, B, C, and D are functions of $\xi \equiv R_p/\lambda_D$. We will reproduce the work from Chen here for the sake of completeness

ABCD are defined in two different regimes. For large ξ ($\xi > 3$),

$$A, B, D(\xi) = a + b(\xi - c)^d \exp[-f(\xi - c)^g]$$
$$C = a + b\exp[-c\ln(\xi - d)] + f(1 - g\ln\xi).$$

This is accurate to within 3%.

Although not as accurate but useful for all ξ, we have

$$A = a + \frac{1}{\frac{1}{b\xi^c} - \frac{1}{d\ln(\xi/f)}}$$
$$B, D = a + b\xi^c \exp\left(-d\xi^f\right)$$
$$C = a + b\xi^{-c}$$

The values for a, b, c, d, f, and g are given in table 3.4.

3.4.3 Effect of electric fields

Electric fields during reconnection can change the average plasma potential all of the tips see which is extracted by fitting the plasma theory to the tip measurements. The effect of these electric fields is non-local because the measured plasma potential is integrated along the path of the probe shaft.

This isn't all though; electric fields can also have a local effect directly measurable by the difference in measurement between tips on one side of the probe vs. the other (see figure 3.3).

		a	b	c	d	f	g
$\xi > 3$	A	1.142	19.027	3.000	1.433	4.164	0.252
	B	0.530	0.970	3.000	1.110	2.120	0.350
	C	0.000	1.000	3.000	1.950	1.270	0.035
	D	0.000	2.650	2.960	0.376	1.940	0.234
$\xi < 3$	A	1.12	.00034	6.87	0.145	110	—
	B	0.50	0.008	1.50	0.180	0.80	—
	C	1.07	0.95	1.01	—	—	—
	D	0.05	1.54	0.30	1.135	0.370	—

Table 3.4: Values for $abcdfg$ in the Chen parametrization of the ion current to a probe.

3.4.3.1 Where does the electric field come from

There are two types of electric field to worry about:

During reconnection we expect the magnetic field to be quickly changing along the axis of the machine as the z directed field reconnects. Since we are dealing with a plasma we expect small static electric fields because of quasineutrality. Also, since the system is toroidally periodic there can be no net static field in the ϕ direction.

3.4.3.1.1 Inductive Fields These are generated by the change in magnetic field as the experiment progresses. Let's discuss what each direction magnetic field change can create:

- z: This is the field that changes most during the experiment because it is reconnecting. Because of toroidal symmetry, the most general closed loop we can construct are annulus arcs. Because of that same symmetry the changing z field will create a negligible r directed electric field but can create toroidal electric fields (in fact the drive coils generate an immense toroidal electric field to drive reconnection).
- r: This field can create either a z or ϕ directed electric field. The ϕ electric field can be generated but once again, because of symmetry,

Figure 3.14: What does an electric field do to the tip measurements diagram.

Figure 3.15: Example plot where E splitting tips

the closed loop on the boundary of a cylinder arc causes E_z to be minimal. One other detriment to E_z is the small magnitudes of B_r which can form in the outflow region of reconnection but are not large.

- ϕ: The source of ϕ field can come from two places: a guide field that is frozen into the plasma (see 1.3.2.3) or the hall fields when the ions and electrons decouple (see 1.2.4). This allows for the formation of both r and z electric fields but because of the often small magnitude of B_ϕ we don't expect too large of fields.

3.4.3.1.2 Static Fields

3.4.3.1.2.1 What is the expected static magnitude

3.4.3.2 What does it do to the tip measurements

3.4.3.2.1 Theoretical understanding

3.4.3.2.2 Physical reality

Figure 3.16: Diagram of local minimums

3.4.4 Minimum measurable density

3.4.4.1 What's the Debye Length

3.4.4.2 Sheath scale

3.4.4.3 Tip-tip sheath overlap

3.5 Curve Fitting

3.5.1 Least squares Fitting

3.5.1.1 Benefits (easy)

3.5.1.2 Issues (not representative of error)

3.5.2 ODR

3.5.2.1 What does error look like in our signals

3.5.2.2 How does ODR take that into account

3.5.2.3 How to apply parameter bounds

3.5.2.3.1 Our method

3.5.2.3.2 Other options

Figure 3.17: Example fit using many shots, multiple local parameter fits, and errors

Figure 3.18: Plot of isat currents from BRL

3.5.3 Monte-carlo Fitting

3.5.3.1 What kind of fits do we get if we just trust a local minimum

3.5.3.2 How do we sample initial conditions

3.5.3.3 More robust methods for getting global minimums

3.5.4 Combining multiple shots

3.5.4.1 Using some time points from each shot

3.5.4.2 Combining all IV points and fitting that

3.5.4.3 Problems

3.5.4.3.1 Timing alignment We'll go into this more in the PA section.

3.5.4.3.2 Variability of plasma parameters

3.6 Results

What is the benefit of doing all this work?

Figure 3.19: Radial chord of initial density with Hutchinson vs BRL.

Figure 3.20: Comparison of expected E from Ohm's law and measured E.

3.6.1 Measurement of low densities

3.6.1.1 How does Hutchinson vs BRL treat ion saturation

3.6.1.2 What are the experimental predictions that Hutchinson vs BRL have

3.6.1.3 How does that change the results

3.6.2 Noise resilience

Not sure how best I should show this.

3.6.3 Fits more accurate

- Each fit uses many data points
    - Don't have to arbitrarily cutoff data points

3.6.4 Extracting electric fields

3.6.4.1 Where do we expect electric fields to be

3.6.4.2 Do we see them there

4 PRESSURE ANISOTROPY PROBE

4.1 How can we measure Pressure Anisotropy

4.1.1 Thomson Scattering

4.1.1.1 Pros

4.1.1.2 Cons

- Requires a lot of playing around with optics
- Size of machine
- Don't have much in-lab expertise
- Getting all the photons is tough
- Don't have optics stuff setup

4.1.2 Directional Langmuir Probe

4.1.2.1 Pros

- Simplicity
- Cost
- Mobility

4.1.2.2 Cons

- Difficult to interpret results precisely (luckily only need relative stuff)
- Can't perfectly model
- Requires a lot of playing around with circuits

Figure 4.1: Picture of feed thru along with drawings of cross-sections.

Figure 4.2: Picture of end of probe

Figure 4.3: CAD diagram of full body

4.2 Hardware

4.2.1 Shaft Components

4.2.1.1 Garages

4.2.1.2 Shaft materials

4.2.1.3 Centering Collet

4.2.1.4 Feed-thru

4.2.2 Tip Components

4.2.2.1 Probe body

4.2.2.1.1 3D printed

4.2.2.1.2 Both sides fit together

4.2.2.2 Tip material + size

4.2.2.2.1 Shell material

4.2.2.2.2 Core material

Figure 4.4: CAD diagram of probe tip

Table 4.1: Length scales of probe

Figure 4.5: Simple circuit diagram of how bias works.

4.2.2.3 Tip Shielding

4.2.2.3.1 Shell Shielding

4.2.2.3.2 Core Shielding

4.2.3 Bdot Coils

Same as Te probe

4.2.4 Connecting Bdot - Probe - Shaft

4.2.5 Length Scales

4.3 Circuitry

4.3.1 Bias Source

4.3.1.1 Principles of operation

4.3.1.1.1 Charging

4.3.1.1.2 Discharging

4.3.1.1.3 Swapping polarity

4.3.1.2 How do we do this

4.3.1.2.1 Cannibalization

Figure 4.6: Full circuit diagram of how bias works.

Figure 4.7: Image of power supply setup

4.3.1.2.2 Discharging

4.3.1.2.3 Safety Diodes for discharging the wrong way.

4.3.1.2.4 Cap arcing

4.3.1.2.4.1 What happened

4.3.1.2.4.2 Solution

4.3.1.2.5 Reducing voltage drop due to inductance

4.3.1.2.5.1 Twisting Wires

4.3.1.2.5.2 Adding cap next to board

4.3.1.2.6 Reversing polarity

4.3.1.2.6.1 Why we break relay

4.3.1.2.6.2 How do we fix this problem

4.3.1.2.7 LabView (ew)

4.3.1.2.7.1 Controls CRIO

4.3.1.2.7.2 Countdown if too high of voltage for swapping

Table 4.2: Bias circuit component values

Table 4.3: Probe circuit component values

Figure 4.8: Schematic of grounding diagram

4.3.1.3 Calibration

4.3.1.3.1 Calibrating power supply output

4.3.1.3.2 Calibrating cap bank output

4.3.2 Probe Circuit

4.3.2.1 Single-ended measurement

4.3.2.1.1 TREX cells

4.3.2.1.2 Why was this bad

4.3.2.1.3 How long did we spend on this :(

4.3.2.2 Differential measurement

4.3.2.2.1 Parallel circuitry

4.3.2.3 Figuring out grounds

4.3.2.3.1 Types of grounds on the ACQ

4.3.2.3.2 Routing grounds

Figure 4.9: Image of swapping box

Figure 4.10: Diagram of swapping

Figure 4.11: Time frame for core movie of IV with separation between relay

4.3.2.4 Swapping current paths

4.3.2.4.1 Why do we want to do this

4.3.2.4.1.1 Takes care of small differences in resistors

4.3.2.4.2 Circuitry to do this

4.3.2.4.2.1 Only swap core stuff

4.3.2.4.2.2 Many relays

4.3.2.4.2.3 Remote control

4.3.2.4.3 How does this change results

4.3.2.5 Circuit Solution

4.3.2.5.1 Solution

4.3.2.5.2 Assumptions

4.3.2.6 Calibration

4.3.2.6.1 Calibrating in isat

Figure 4.12: Circuit diagram of probe circuitry

Figure 4.13: Image of probe circuitry

Figure 4.14: Probe signals showing coupling

4.3.2.6.2 Calibrating on the exponential

4.3.2.6.3 Rotation calibration

4.3.2.7 The path to ground

4.3.2.7.1 Why do we have to care

4.3.2.7.2 How do we set it

4.3.2.8 Capacitive coupling

4.3.2.8.1 Where does it come from

4.3.2.8.2 Where do we see it

4.3.2.8.3 Magnitude of effect

4.3.2.8.4 Removing effect

Figure 4.15: Figure of theoretial plasma distributions and JV curves

4.4 Plasma Theory

4.4.1 A full model

4.4.1.1 Plasma Distribution

4.4.1.2 Ion particle flux

4.4.1.2.1 Mach effects

4.4.1.3 Electron particle flux

4.4.1.3.1 Minimum velocity

4.4.1.3.2 Area

4.4.1.3.3 Particle Flux

4.4.1.4 Theoretical JV curves

4.4.1.4.1 Reading radar plots

4.4.1.4.1.1 Radar axis

4.4.1.4.1.2 Solutions

4.4.2 Individual tip fitting

4.4.2.1 Why do this instead

4.4.2.2 Methods

4.4.2.2.1 Aforementioned methods

Figure 4.16: Simple drawing of what might happen to ions

4.4.2.2.2 Exponential

4.4.2.2.2.1 Ions don't hit core

4.4.2.2.2.2 Take care of this by allowing free parameters

4.4.2.2.2.3 Converting from plasma parameters to exponential

4.4.2.2.2.4 Converting back

4.4.2.2.2.5 Taking care of errors

4.4.2.3 Extracting plasma parameters

4.4.2.3.1 Spherical harmonics

4.4.2.3.2 Error bars

4.4.2.3.2.1 Monte-carlo

4.4.2.3.2.2 Other options

4.4.2.3.2.3 Why can we ignore non-bi-maxwellian features

4.5 Curve Fitting

We will use ODR like mentioned in Te section.

Figure 4.17: Comparison of alignment using Bz = 0 or bdot cross correlation

4.5.1 Combining multiple shots

4.5.1.1 Timing alignment

4.5.1.1.1 Bz = 0 alignment

4.5.1.1.1.1 Aligns where we expect Anisotropy

4.5.1.1.1.2 Depends heavily on prior bdot data

4.5.1.1.2 Bdot cross correlation

4.5.1.1.2.1 Still requires smoothing to get rid of initial oscillation impact

4.5.1.1.3 Result comparison

4.5.1.2 Averaging of core data

4.5.1.2.1 Weighted average of relay on/off

4.5.2 Fitting shell

4.5.2.1 Use all shell data

4.5.2.2 Fit using BRL

4.5.2.3 Fits pretty good

4.5.3 Fitting core

4.5.3.1 Fitting using full routine

4.5.3.1.1 Slow

4.5.3.1.2 Model not perfectly representative

4.5.3.2 Fitting tips independently

4.5.3.2.1 Individual fitting method

4.5.3.2.2 Holding Vprobe constant

4.5.3.2.3 Exponential fitting method

4.5.3.2.4 Spherical harmonics

4.6 Results

Figure out what my results are once I have some reasonable results

4.6.1 Drive cylinder

4.6.1.1 What did we try

4.6.1.2 Why didn't we see pressure anisotropy

4.6.1.2.1 In the bdot data

4.6.1.2.2 In the PA data

4.6.2 3 coil

4.6.2.1 How much anisotropy should we expect

4.6.2.2 What setups did we try

4.6.2.3 Old results of just comparing radar plots

4.6.2.4 New results using independent fitting

5 CONCLUSION

5.1 Summary of thesis

5.1.1 Using pressure waves to measure density

5.1.2 Upgrading langmuir probe analysis and new possibilities

5.1.3 Using directional langmuir probes for measuring pressure anisotropy

5.2 Future work

5.2.1 Pressure wave

5.2.1.1 Aforementioned future work

5.2.1.2 Extracting cross-field diffusion

5.2.2 Langmuir Probe

5.2.2.1 Prepared to analyze lots of cool data

5.2.2.2 Extract out electric field

5.2.3 PA Probe

5.2.3.1 Try more configurations

5.2.3.2 Map out full pressure anisotropy during shot

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